

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	01 October 2026
Team ID	SWTID-2026-1665
Project Name	EduGenie – Google Gemini Powerful Learning Assistant
Maximum Marks	4 Marks

Technical Architecture

EduGenie follows a layered, cloud-based architecture. Students and teachers interact with a React web interface, which communicates with a Flask REST API. The API builds learning-focused prompts and calls the Google Gemini API to generate explanations, summaries and quizzes, while Firebase Authentication, Cloud Firestore and Cloud Storage handle sign-in, user data and uploaded files. The whole application is containerised and deployed on Google Cloud Run.

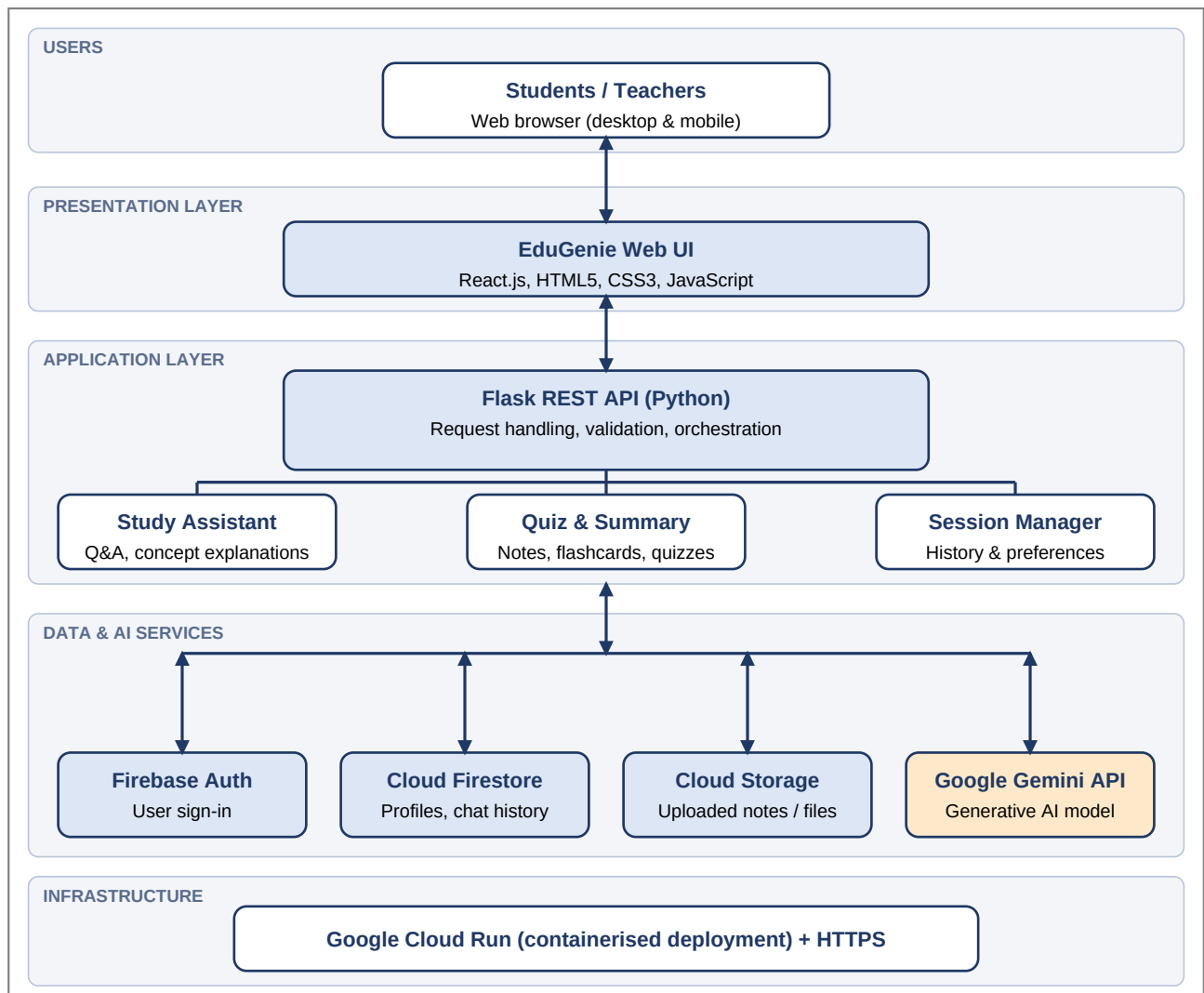


Figure 1: Technical Architecture of EduGenie

Table-1: Technical Architecture Components

S.No	Component	Description	Technology
1	User Interface	Web interface through which students and teachers chat with EduGenie, upload notes, and view explanations, summaries and quizzes.	HTML5, CSS3, JavaScript, React.js
2	Application Logic-1	Core backend service that receives requests, validates input and routes them to the proper learning module.	Python, Flask (REST API)
3	Application Logic-2	Prompt-engineering layer that builds subject-aware prompts for explanations, summaries, flashcards and quizzes.	Python (prompt templates)
4	Application Logic-3	Session and learning-history management, including user preferences and conversation context.	Python, Flask sessions
5	Database	Stores user profiles, chat history, generated study material and quiz results.	Cloud Firestore (NoSQL)
6	Cloud Database	Managed, real-time, cloud-hosted database with automatic scaling and backup.	Google Firebase / Cloud Firestore
7	File Storage	Storage for uploaded notes, documents and generated study files.	Google Cloud Storage
8	External API-1	Generative AI API used for understanding questions and producing learning content.	Google Gemini API
9	External API-2	Authentication service for secure sign-up and login.	Firebase Authentication
10	Machine Learning Model	Large language model that performs text understanding, reasoning, summarisation and quiz generation.	Google Gemini (LLM)
11	Infrastructure (Application & Database)	Containerised deployment of the application on a managed cloud platform with HTTPS.	Docker, Google Cloud Run, Git/GitHub

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open-source frameworks used to build the front end and back end of the application.	React.js, Flask, Python
2	Security Implementations	User authentication, API keys held in environment variables, input validation, HTTPS encryption and role-based access to user data.	Firebase Auth, HTTPS/TLS, Firestore security rules, environment variables
3	Scalable Architecture	Three-tier, stateless design: presentation, application and data/AI layers scale independently.	Cloud Run auto-scaling, Firestore
4	Availability	Managed cloud services with automatic scaling and restart, giving high availability across regions.	Google Cloud Run, Firebase
5	Performance	Concise prompts, response streaming and caching of repeated queries keep response times low.	Gemini API, caching, async requests